

ClimateCraft — Melting Ice Trackers

Complete Project Definition & Pipeline Planning Document

Purpose of this document: Define exactly what the project is, why it is useful, who uses it, what the system should produce, and how the major components connect. This document is intended to be the foundation from which the implementation pipeline can be designed.

1. Executive Summary

Melting Ice Trackers is a geospatial AI-based glacier monitoring system. Its purpose is not simply to calculate an ice-loss number. The system converts satellite observations of glaciers into structured information about **where ice exists, how much ice-covered area exists, how that area changes over time, where the change occurs, and which changes deserve further investigation.**

The central idea is: **Satellite observation → ice identification → ice extent measurement → temporal comparison → change detection → prioritization/visualization → expert investigation.**

The system should be treated as a **monitoring and decision-support system**, not as an autonomous scientific or policy decision-maker. It provides measurements and flags that researchers, environmental organizations, and disaster-risk teams can combine with other information.

2. Problem Statement

Glaciers are large, geographically distributed, and continuously changing. Satellite imagery provides a way to observe them repeatedly, but raw imagery is difficult to turn into consistent measurements when many glaciers and many dates are involved.

Problem: There is a need for an automated system that can transform repeated satellite observations into quantitative and spatial information about glacier ice extent and its change over time.

A weak version of the project would only answer: *“How many square kilometres of ice were lost?”* The stronger project answers: *“Which glacier changed, how much did it change, where did the change occur, how quickly did it change, is the change unusual relative to its history, and which locations should experts look at more closely?”*

3. Why the Project Is Useful

Capability	What it provides	Why it matters
Measurement	Ice-covered area for a glacier/date	Creates a quantitative observation rather than a subjective visual judgment.
Historical tracking	Measurements across multiple dates	Shows whether change is gradual, stable, or unusually rapid.
Spatial change	Location of changed ice regions	Tells experts where to investigate rather than giving only one total number.
Large-scale monitoring	Same workflow across many glaciers	Reduces the need for manual inspection of every image.
Prioritization	Flags notable or unusual changes	Helps experts focus attention on locations requiring closer analysis.
Research support	Structured historical observations	Can support glacier, climate, remote-sensing, and hydrological research.

4. Who Will Use It?

Glaciologists: Study glacier evolution, spatial retreat, and long-term changes.

Climate researchers: Combine glacier observations with temperature, precipitation, and other environmental variables.

Remote-sensing researchers: Use automated satellite-image analysis and derived measurements.

Environmental / government agencies: Monitor many glaciers and identify locations requiring closer assessment.

Disaster-risk researchers and agencies: Use glacier and related environmental change as one input to broader hazard assessment.

Students / educators / public: Explore glacier change through maps, historical graphs, and visual comparisons.

5. What the System Actually Does

The system revolves around five core questions:

1. WHERE?	Which pixels/regions of the observation correspond to glacier ice?
2. HOW MUCH?	What is the ice-covered geographic area?
3. WHEN?	At what date was that observation made?
4. HOW DID IT CHANGE?	How does the ice extent differ from earlier observations?
5. WHAT DESERVES ATTENTION?	Any observed change significant or unusual enough to be flagged for expert investigation?

6. Example: What Happens to an 8 km² Loss Value?

Suppose the system reports that Glacier X decreased from 100 km² to 92 km² of measured ice-covered area between two observation periods. The raw result is **8 km² decrease**. That number becomes useful when the system adds context:

Initial extent	100 km²
Current extent	92 km²
Absolute change	-8 km²
Percentage change	-8%
Observation period	2020–2025 (example)
Historical rate	Compare with earlier periods
Spatial location	Show where the changed area occurred
Change status	Normal / notable / significant according to defined criteria

The system therefore does not stop at “8 km²”. It can turn the measurement into a monitoring record and a spatially explicit flag. An expert can then investigate the glacier using additional scientific information. The system should **not claim that an observed area decrease alone proves a specific cause**, because snow cover, seasonality, image conditions, ice dynamics, and measurement uncertainty can influence observations.

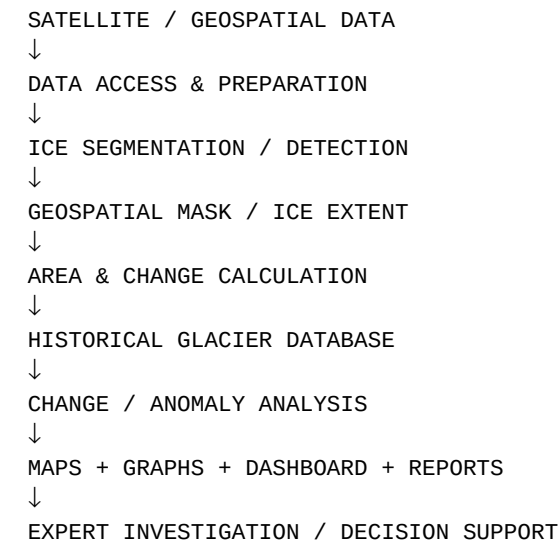
7. End-to-End Conceptual Pipeline

This is the main pipeline that the eventual implementation should realize. It is deliberately expressed at the system level rather than as a list of coding tasks.

1. Observation	Receive/select a satellite observation associated with a glacier and date.
2. Image understanding	Determine the spatial region corresponding to glacier ice.
3. Ice representation	Represent the detected ice spatially so that its extent can be measured.
4. Geospatial measurement	Convert the detected spatial extent into real-world area using the image's geospatial properties.
5. Historical record	Store the measurement together with glacier identity, date, and relevant metadata.
6. Change analysis	Compare current and previous observations to calculate absolute change, percentage change, and rate of change.
7. Spatial change detection	Determine where the glacier's observed ice extent changed.
8. Significance / anomaly assessment	Compare the new observation with the glacier's historical behavior and flag notable changes using defined thresholds.
9. Visualization	Present maps, before/after imagery, historical trends, statistics, and flagged locations.
10. Expert follow-up	Allow researchers or agencies to investigate flagged glaciers using additional environmental and scientific data.

8. System Architecture at a Conceptual Level

The following layers describe the eventual system without committing to a specific programming language or model:



9. What Information the System Needs Conceptually

Satellite observation	The image/observation from which glacier extent is determined.
Date/time	Required to compare observations correctly.
Glacier identity	A stable identifier/name for the glacier being monitored.
Geospatial reference	Information connecting image pixels to real-world geographic location.
Spatial resolution	Needed to translate detected pixels into geographic area.
Ice extent	The spatial representation of detected glacier ice.

Calculated area	The real-world area derived from the ice extent.
Historical measurements	Previous observations for the same glacier.
Change metrics	Absolute change, percentage change, rate, and spatial change.
Quality/uncertainty information	Information needed to avoid presenting uncertain observations as exact facts.

10. What the Final User Should See

The user interface should make the scientific story understandable without requiring the user to inspect raw imagery.

GLACIER SELECTION

→ Select glacier / region / date range

CURRENT STATUS

→ Current ice-covered area
→ Absolute change
→ Percentage change
→ Change rate where supported

SPATIAL VIEW

→ Current ice extent on a map
→ Previous extent
→ Highlighted changed regions

HISTORICAL VIEW

→ Ice extent vs. time graph
→ Historical measurements

MONITORING FLAG

→ Normal / notable / significant according to transparent criteria
→ Explanation of why it was flagged

INVESTIGATION

→ Additional context for researchers

11. What the System Should NOT Claim

- An observed reduction in ice-covered area is not automatically identical to a measured volume of ice lost.
- An area change by itself does not prove why the change occurred.
- A detected change does not automatically mean a disaster will happen.
- A monitoring flag is not a scientific or government decision.
- Forecasts, if later added, are estimates based on assumptions and historical patterns, not guarantees.

12. Change Analysis the Project Should Support

Metric	Meaning
Absolute area change	Current measured ice-covered area minus reference area.
Percentage change	Change relative to the reference ice-covered area.
Rate of change	Change normalized by the observation period when appropriate.
Spatial change	Which geographic regions gained/lost observed ice extent.

Metric	Meaning
Historical deviation	How the latest observation compares with the glacier's historical behavior.

13. Suggested Team-Level Responsibilities

AI / Computer Vision	Image understanding and glacier-ice segmentation; evaluation of the quality of ice detection.
AI / Analytics	Geospatial area calculation, temporal change analysis, anomaly/significance logic, and optional forecasting.
Cyber / Security	Authentication, authorization, secure data handling, API security, upload validation, and application security.
Cyber / Infrastructure	Secure deployment, logging, monitoring, integrity controls, and security testing.

14. Recommended Development Pipeline

This is the order in which the project should eventually be designed and implemented. It is intentionally different from the conceptual explanation above: this section turns the concept into development stages.

Phase 1 — Define the monitoring unit	Decide what constitutes a glacier, observation, date, region, and monitoring record.
Phase 2 — Define the output contract	Specify exactly what one processed observation must return: ice extent, area, metadata, quality indicators, etc.
Phase 3 — Design the geospatial workflow	Determine how imagery, coordinates, resolution, projection, and glacier boundaries will be handled.
Phase 4 — Design ice-detection/segmentation	Obtain and evaluate the computer-vision approach that converts an observation into an ice extent.
Phase 5 — Build measurement logic	Convert the spatial ice result into consistent real-world measurements.
Phase 6 — Build temporal tracking	Store and compare observations for each glacier across dates.
Phase 7 — Build change intelligence	Implement change metrics, spatial change detection, and transparent significance/flagging rules.
Phase 8 — Build visualization	Create glacier pages, maps, before/after comparison, historical graphs, and monitoring status.
Phase 9 — Add security	Protect data, APIs, users, uploads, and deployment infrastructure.
Phase 10 — Validate end-to-end	Test whether the system's measurements and change results are accurate and scientifically defensible.

15. What Would Make the Project Successful?

- ✓ A user can select a glacier and a time period.
- ✓ The system can identify the glacier's ice extent from satellite observations.
- ✓ The system can calculate a geographically meaningful ice-covered area.
- ✓ The system can compare observations across dates.
- ✓ The system can show where spatial change occurred.
- ✓ The system can distinguish ordinary change from predefined notable/significant change criteria.
- ✓ The dashboard makes the result understandable to a non-technical user.
- ✓ Experts can access the underlying evidence and metadata rather than seeing only a single score.
- ✓ The system clearly communicates uncertainty and limitations.

16. Final Project Definition

Melting Ice Trackers is an intelligent glacier-monitoring and change-detection platform that uses satellite observations to identify glacier ice extent, quantify its geographic area, track changes across time, locate spatial changes, and flag notable changes for expert investigation.

The key output is therefore not simply an “ice-loss number.” The key output is a **structured monitoring record** containing the glacier, observation date, measured ice extent, area, historical comparison, spatial change, and monitoring status. This makes the system useful as a research and decision-support layer.

17. One-Line Mental Model

Observe → Identify ice → Measure → Compare → Detect change → Locate change → Flag → Visualize → Investigate.

This document intentionally focuses on the **project itself and the pipeline it needs to achieve**. Specific datasets, annotation/mask creation, model architecture, training strategy, software stack, and deployment can be designed as separate engineering decisions after the project definition is finalized.